Python Theory Questions

What is Python?

– Python is a High-level programming language & General-purpose programming language known for its simple, readable syntax. It supports multiple programming paradigms like procedural,object-oriented, functional and is used in web development,automation,data science & AIML.

What are the benefits of using python language?

–Easy to learn and read, open source, platform independent, python is versatile, supports oop and functional programming.

Is Python a compiled language or an interpreted language ?

—Python is interpreted language, code is compiled to byte code first and then executed by the python virtual machine.

What does the "#" symbol do in the python ?

–"#" It denotes a single-line comment. Anything after # on that line is ignored to execute and used for documentation.

Mutable Vs Immutable Data Types?

–Mutable :can be modified after creation.
List, Dictionary,Set
–Immutable: Cannot be modified after creation.
String, Tuple, Integer, Float

Difference Between Set and Dictionary ?

– Set stores unique values, no keys, Uses{1,2,3}, Its an unordered collection of unique values.
– Stores key-value pairs, key must be unique, Its an unordered collection of key-value pairs.

List Comprehension with Example ?

– A short way to create lists in single line instead of a for-loop.

Lambda Function

– An anonymous function to written in one line.

What is Pass in the Python ?

– A null Statement It does nothing. Used as aPlaceholder where Python's Syntax requires a statement, like an empty function body.

Difference Between / and // ?

– / is true Division and always returns a float. // is floor division it divides and rounds down to the nearest integer.

Exception Handling in Python ?

– exception Handling is used to manage runtime errors without crashing the program. Python provides try,except,else,and finally blocks to handle exceptions gracefully.

Swapcase() Function ?

– The Swapcase() method coverts all uppercase letters into lowercase and all lowercase letters into uppercase.

For loop VS while loop

– A for loop is used when the number of iterations is known beforehand or when iterating over a sequence.

– A while loop is used when the number of iterations is unknown and the exception depends on the condition.

Passing a function as an argument

– Functions are treated as first-class objects. So a function can be passed as an argument to another function,returned fro another function, or assigned to a variable.

***args and **kwargs**

– *args allows a function to accept any number of positional arguments" as tuple.

– **kwargs allows a function to accept any number of "Keyword arguments" as Dictionary.

Scope in Python

– It determines where a variable can be accessed. Python follows LEGB Rule-Local(Inside the current function),Enclosing(Inside an outer function), Global(Defined outside all functions), Built-in(Names provided by python itself).

Dynamically Typed Language

– Python is dynamically typed, we don't need to declare data type of a variable. The type is determined automatically during execution and can change during runtime.

Break, Continue, and Pass

– Immediately exits the loop, Continue- skips the current iteration and moves to the next one.
Pass- it does nothing acts as a place holder.

Built- in Data Type

– Numeric(int,float,complex),Boolean(boll), String(str), List, Tuple, Set, Dictionary, Bytes

Flooring a number

– Flooring uses // floor division for two numbers, it means rounding a number down to nearest smaller integer. Ex:4.9 =4.

Key Features of Python

– Simple and readable syntax, Open source, Platform independent, object-oriented, supports multiple program paradigms, Dynamically typed.

= = VS is operator.

– = = operator checks whether two values are equal, “is” operator checks whether two variables refers to the same object in memory.

How Python Functions Work ?

– A Function is a reusable block of code defined using the def keyword. Functions execute only when called and can accept parameters and return values.

Is Indentation Necessary ?

– Yes, Indentation is mandatory in python because it defines blocks of code . Incorrect indentation results in an Indentation error.

Capitalize the First Letter of a String?

– We uses Captalize() method, which converts the first character to uppercase and the remaining characters to lowercase.

Writing Comments

– Comments re used to explain the code, single- line comments use #, Multi-line comments are commonly written using triple quotes (‘”) or(“ “ “) ,though these are technically multi-line strings.

[::-1] Usage

– [::-1] is a slicing technique used to reverse a sequence such as a string, list, or tuple.

What are DocStrings ?

– These are the documentation strings used to describe modulus, classes, and functions. They help explain the purpose, parameters, and return values , making the code easier to understand and maintain.__doc__ and help() function.

Slicing in Python

– Slicing is the process of extracting a portion of a sequence such as string, list, or tuple by specifying a start index, and optional step value.sequence[start:stop:step] It works on springs, lists, tuples and supports negative indexes.

Difference Between NumPy and SciPy

– NumPy is mainly used for fast numerical computations and efficient array operations.

– SciPy is built on top of NumPy and provides advanced scientific computing functions such as optimization , Statistics, signal processing, and linear algebra.

Map()Function

– The map() function applies a specified function to every element of an iterable and returns a new iterable containing the transformed values.

Ways to reverse a String

– Slicing with [::-1] ,the reversed() function, loops, recursion, or by joining reversed characters, Slicing is the most common and efficient method.

Need for Break Statement

– The Break statement is used to terminate a loop immediately when a specific condition is met, avoiding unnecessary iterations.

Difference between map() and lambda()

– Map() is built-in function used to apply a function to every element of an iterable.

–Lambda() is a way to define that function inline without a formal def, it si often used together with map() for short, one-time operations.

Limitations of Python

–Slower execution compared to compiled languages like C++ ,the Global Interpreter Lock restricts true parallelism for CPU-bond multithreading, higher memory usage, and it's not ideal for mobile app development.

Abstract Classes and uses

– A class that cannot be instantained directly . It contains one or more abstract methods that must be implemented by its child classes. Abstract Classes help enforce a common structure across related classes.

Valid Dictionary Keys

– Dictionary Keys must be unique, immutable, and Hashable. Valid Keys include integers, strings, floats, and tuples,Lists, dictionaries, and sets cannot be used as dict keys because they are mutable.

Declare Private Variables

–Private variables are declared a single underscore prefix(_var) signals "protected", and a double underscore(__var) triggers name mangling, making it less accessible from the outside of the class.

Difference Between Array and List

– An Array stores elements of the same data type and is more memory-efficient for numerical operations.

– A list can store elements of different data types and is more flexible, making it the most commonly used collection type in python.

Iterators And Generators

– An iteretaor is an object that allows sequential traversal of elements using the iterator protocol.

— A generator is a special type of iterator created using the yield keyword .It generates values one at a time, making it highly memory-efficient.

Accessing parent class Attributes

– A child class can access parent class attributes through inheritance. Python also provides the super() function to access parent class methods and attributes conveniently.

Multiprocessing and Multithreading

– Multithreading uses multiple threads within the same process. Threads share memory and are suitable for I/O-bound tasks.

–Multiprocessing creates separate processes with independent memory spaces. It is ideal for CPU- intensive tasks and archives true parallel execution.

Merging Dictionaries

– Dictionaries can be merged using methods like update(), dictionary unpacking (**), or the merge operator (|) .If keys exist ,the later value overwrites the earlier one.

Overloading Constructors

–No. Python does not support constructor overloading directly.

Similar functionally can be achieved using default parameter values or variable-length arguments(*args and **kwargs).

Filter with lambda() expression

– The filter() function is used to select elements from an iterable that satisfy a condition. A lambda expression is used to define that condition in a concise manner. It returns only the elements for which the condition evaluates to true.

Intermediate-Level Theory Questions

How is Memory allocation and Garbage collection handled in python ?

– Python uses automatic memory management. Objects are stored in the python Heap, and memory is managed through reference counting and garbage collection. When an object is no longer referenced,its memory is automatically released.

What is the Python switch Statement ?

– Python did not support switch-statements. Instead this we use if-elif-else or dictionaries. The match-case statement provides switch- like functionality.

Explain Closures in Python ?

– A Clouser is an inner function that remembers and accesses variables from its outer function,even after the outer function has completed execution.

Function Annotations

– Function annotations are optional type hints that describe the expected input and output types of a function. They improve readability and support static analysis but are not enforced by Python.

__init__() in Python

– The__init__() method is known as the construct in python. It is automatically called when an object is created and is used to intialize object attributes.

NameSpace in Python

– A namespace is a mapping between names and objects. Python searches for variables using the LEGB rule: Local,Enclosing, global, and Built-in.

Slicing in Python

– Slicing is a technique used to extract a specific portion of a sequence such as a string, list, or tuple by specifying the start, stop, and optional step values.

Memory Management in Python

– Python uses automatic memory management through the python Heap, reference counting, and a garbage collector to efficiently allocate and free memory.

Data Abstraction

– Data abstraction hides implementation details and exposes only the necessary functionality to user, making programs simpler and more secure.

Encapsulation

– Encapsulation is the process of combining data and methods into a single class while restricting direct access to sensitive data.

Polymorphism

— polymorphism allows the same method or interface to perform different actions depending on the object or class.

Does Python Support Multiple Inheritance ?

– Yes. Python Supports multiple inheritance, allowing a class to inherit from multiple parent classes. It uses the method Resolution Order(MRO) to determine which parent method should be called.

Generators VS Iterators

– Both generators and iterators are used for iteration. Generates are more memory-efficient because they generate values on demand instead of storing all values in memory.

How to Debug a Python Program ?

– Python programs can be debugged using print statements, the built-in pdb debugger, logging, or IDE debugging tools such as PyCharm and VS Code.

Decorators

– A decorator is a function that adds additional functionality to another function without modifying its original implementation.

Sorting Technique is Used by sort() and sorted()

– Python's sort(0 and sorted() functions use Timsort, a stable hybrid sorting algorithm derived from merge sort and insertion sort.

Shallow Copy VS Deep Copy

– A shallow copy copies only the top-level object , while a deep copy creates completely independent copies of all nested objects.

List VS Tuple

– Lists are mutable collections suitable for data that changes, whereas tuples are immutable collections that are faster and more memory-efficient.

Is Tuple Comprehension Supported ? Why?/ Why not?

– Python does not support tuple comprehensions. Expressions inside parentheses create generators because generators are more memory-efficient.A tuple can then be created by passing the generator to the tuple() function.

Dictionary Comprehension with Example

–Dictionary comprehension is a compact way to create dictionaries by generating key- value pairs in a single expression ,making the code shorter and easier to read.

Monkey Patching

– Monkey Patching is the practice of dynamically modifying or replacing methods or attributes of existing classes or objects at runtime.

Staticmethod VS Classmethod

– A Static method does not depend on either the instance or the class, while a class method works with the class itself and can access or modify class-level data.

List Comprehension Vs For Loop Speed

– List Comprehensions are usually faster and more concise than for loops because they are optimized internally. However, for complex operations, a loop may provide better readability and flexibility.